RM520N–GL / RTL8125BG / independent US915 OOB prototype

48 V passive cable power –> protected buck 12 V –> modem / Ethernet rails

Auxiliary DC, USB, modem coax and mounting share system GND; no power galvanic isolation.

Supplier acceptance and hardware tests: docs/validation.md.

Passive cable–power connections and protected auxiliary DC

File: power–input.kicad_sch

Protected 12 V bus and regulated low–voltage rails

File: power–rails.kicad_sch

RM520N–GL primary / RM551E–GL compatible M.2 socket, SIM 1, eUICC (SIM 2), USB–C

File: modem–interfaces.kicad_sch

RTL8125BG 2.5 GbE

File: ethernet.kicad_sch

48 V passive input protection and 12 V synchronous buck

File: class6–converter.kicad_sch

Modem current ceiling and always–on supply

File: class6–grants.kicad_sch

OOB controller, US915 radio and recovery interface

File: oob–controller–radio.kicad_sch

OOB monitoring, power control and modem recovery

File: oob–management.kicad_sch

OOB USB ownership and reset fallback

File: oob–usb–ownership.kicad_sch

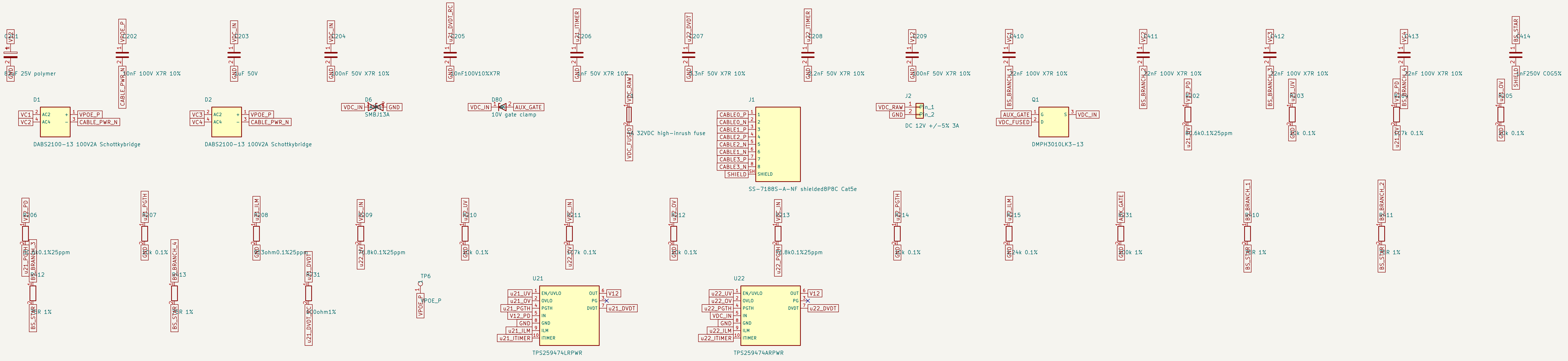
Validation: docs/validation.md; BOM and build files: manufacturing/rev–a/
ENGINEERING PROTOTYPE – not a fabrication approval

Sheet: /
File: carrier.kicad_sch

Title: RM520N–GL passive carrier with independent US915 OOB

Size: A2	Date: 2026–10–03	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/10

Passive cable−power connections and protected auxiliary DC



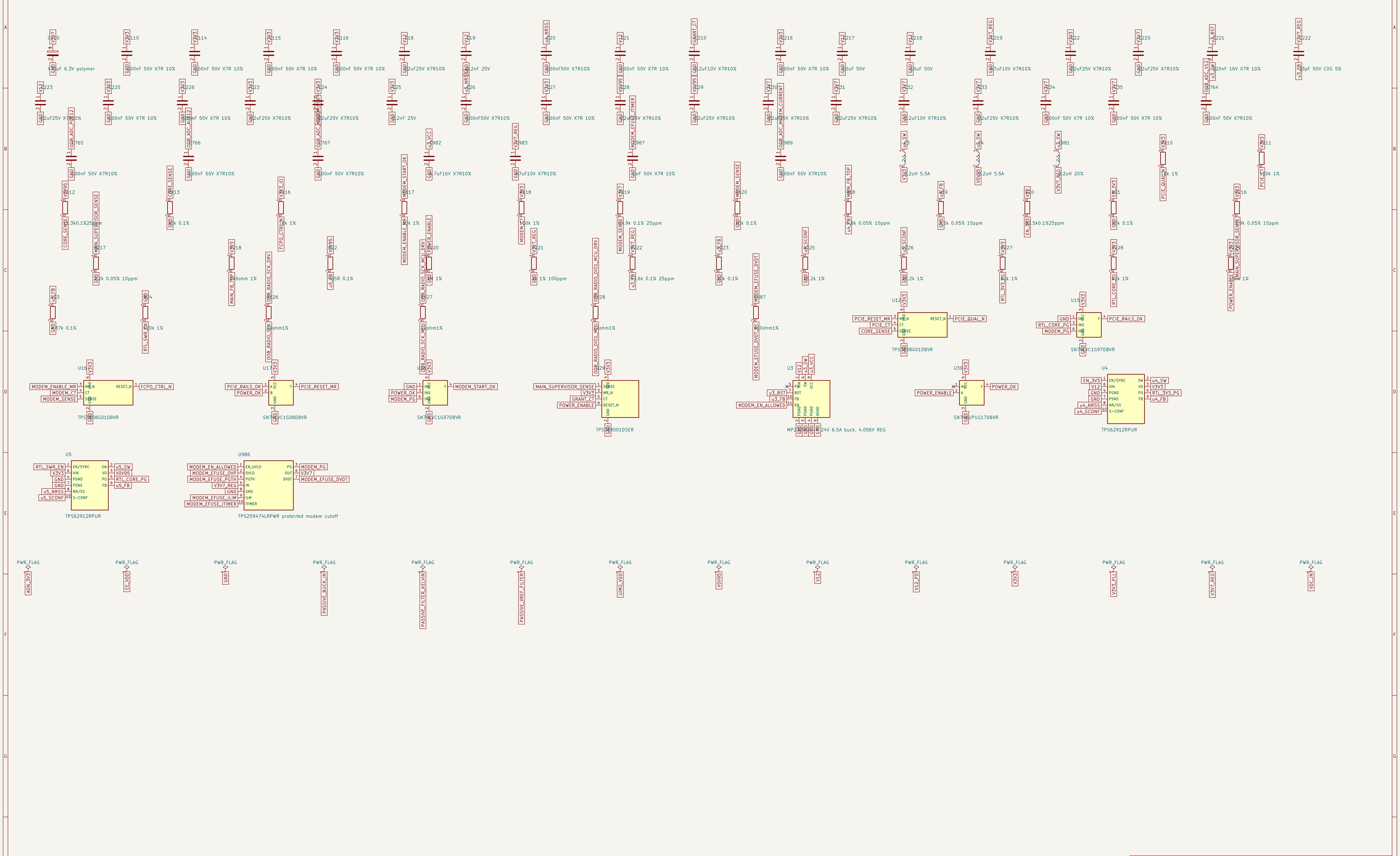
Global labels link sheets; cable return reaches system GND through explicit CMC
Rev A build preparation − current source/native release checks required

Sheet: /Passive cable−power connections and protected auxiliary DC/
File: power−input.kicad_sch

Title: Passive cable−power connections and protected auxiliary DC

Size: A2
Date: 2026−10−03
Rev: A
KiCad E.D.A. 10.0.6
Id: 2/10

Protected 12 V bus and regulated low-voltage rails



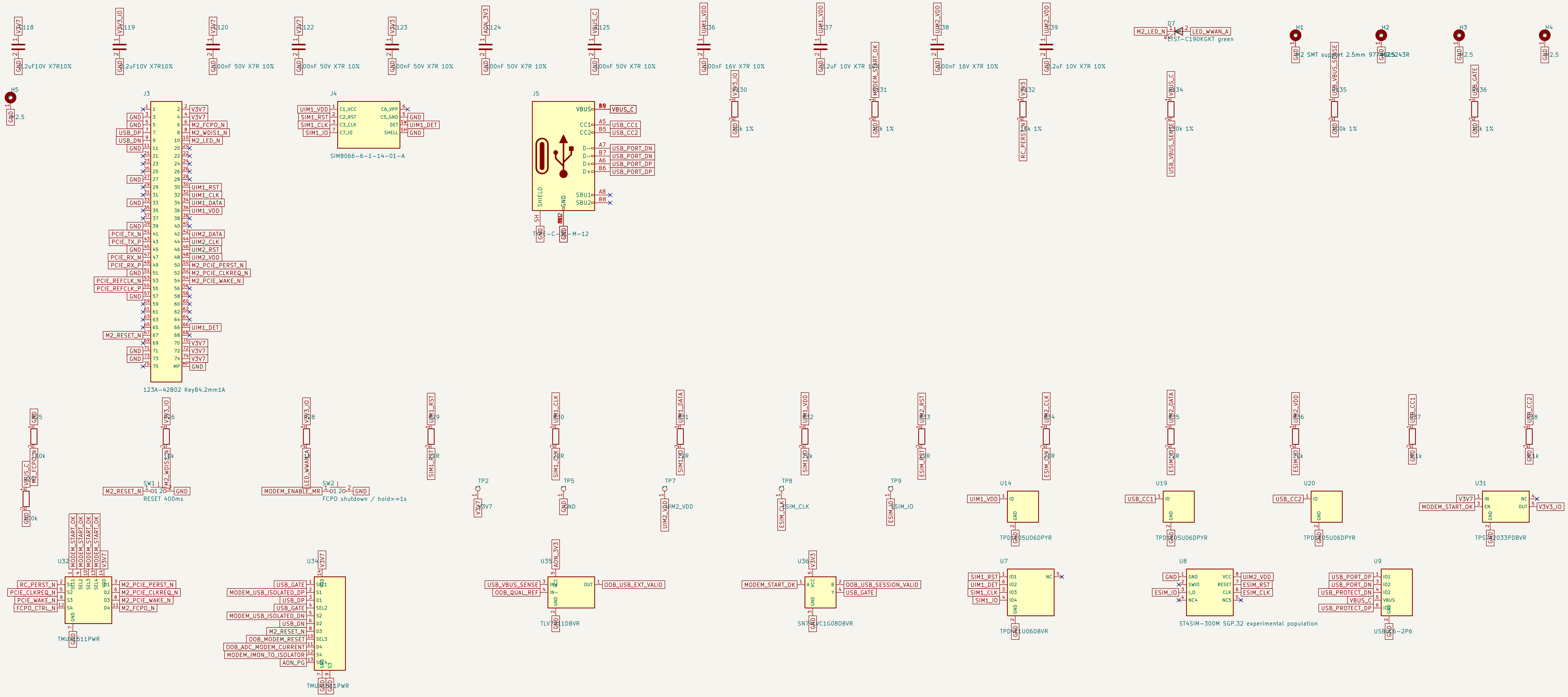
Global labels link sheets; cable return reaches system GND through explicit CMC
Rev A build preparation — current source/native release checks required

Sheet: /Protected 12 V bus and regulated low-voltage rails/
File: power-rails.kicad_sch

Title: Protected 12 V bus and regulated low-voltage rails

Size: A2 Date: 2026-10-03 Rev: A
KiCad E.D.A. 10.0.6 Id: 3/10

RM520N-GL primary / RM551E-GL compatible M.2 socket, SIM 1, eUICC (SIM 2), USB-C


C118 V3V7 GND 2.2uF10V X7R10%
C119 V3V3_IO GND 2.2uF10V X7R10%
C120 V3V7 GND 100nF 50V X7R 10%
C122 V3V7 GND 100nF 50V X7R 10%
C123 V3V3 GND 100nF 50V X7R 10%
C124 AON_3V3 GND 100nF 50V X7R 10%
C125 VBUS_C GND 100nF 50V X7R 10%
C36 UIM1_VDD GND 100nF 16V X7R 10%
C37 UIM1_VDD GND 2.2uF 10V X7R 10%
C38 UIM2_VDD GND 100nF 16V X7R 10%
C39 UIM2_VDD GND 2.2uF 10V X7R 10%
D7 M2_LED_N LED_WWAN_A LTST-C190KGKT green
H1 GND M2 SMT support 2.5mm 9774025243R
H2 GND
H3 GND M2.5
H4 GND M2.5
H5 GND M2.5
R130 V3V3_IO GND 10k 1%
R131 MODEM_START_OK GND 10k 1%
R132 V3V3 RC_PERST_N 10k 1%
R134 VBUS_C USB_VBUS_SENSE 100k 1%
R135 USB_VBUS_SENSE GND 100k 1%
R136 USB_GATE GND 10k 1%
J3 123A-42B02 KeyB4.2mm1A
GND GND USB_DP USB_DN GND GND GND GND PCIE_TX_N PCIE_TX_P GND PCIE_RX_N PCIE_RX_P GND PCIE_REFCLK_N PCIE_REFCLK_P GND M2_RESET_N GND GND
V3V7 V3V7 M2_FCPO_N M2_WDIS1_N M2_LED_N UIM1_RST UIM1_CLK UIM1_DATA UIM1_VDD UIM2_DATA UIM2_CLK UIM2_RST UIM2_VDD M2_PCIE_PERST_N M2_PCIE_CLKREQ_N M2_PCIE_WAKE_N UIM1_DET V3V7 V3V7 V3V7 GND
J4 SIM8066-6-1-14-01-A
UIM1_VDD C1_VCC
SIM1_RST C2_RST
SIM1_CLK C3_CLK
SIM1_IO C7_IO
C6_VPP
C5_GND GND
DET SW UIM1_DET
SHELL SH GND
J5 TYPE-C-31-M-12
VBUS VBUS_C
CC1 A5 USB_CC1
CC2 B5 USB_CC2
D- A7 USB_PORT_DN
D- B7 USB_PORT_DN
D+ A6 USB_PORT_DP
D+ B6 USB_PORT_DP
SBU1 A8
SBU2 B8
SHIELD SH GND
GND GND
R25 GND M2_FCPO_N 100k
R? VBUS_C GND 100k
R26 V3V3_IO M2_WDIS1_N 10k
R28 V3V3_IO LED_WWAN_A 1k
SW1 M2_RESET_N GND RESET 400ms
SW2 MODEM_ENABLE_MR GND FCPO shutdown / hold>=1s
R29 UIM1_RST SIM1_RST 22R
R30 UIM1_CLK SIM1_CLK 22R
R31 UIM1_DATA SIM1_IO 22R
R32 UIM1_VDD SIM1_IO 20k
R33 UIM2_RST ESIM_RST 22R
R34 UIM2_CLK ESIM_CLK 22R
R35 UIM2_DATA ESIM_IO 22R
R36 UIM2_VDD ESIM_IO 20k
R37 USB_CC1 GND 5.1k
R38 USB_CC2 GND 5.1k
TP2 V3V7
TP5 GND
TP7 UIM2_VDD
TP8 ESIM_CLK
TP9 ESIM_IO
U14 UIM1_VDD IO GND TPD1E05U06DPYR
U19 USB_CC1 IO GND TPD1E05U06DPYR
U20 USB_CC2 IO GND TPD1E05U06DPYR
U31 TPS7A2033PDBVR
V3V7 IN
MODEM_START_OK EN
NC
OUT V3V3_IO
U32 TMUX1511PWR
MODEM_START_OK MODEM_START_OK MODEM_START_OK MODEM_START_OK V3V7
RC_PERST_N S1 D1 M2_PCIE_PERST_N
PCIE_CLKREQ_N S2 D2 M2_PCIE_CLKREQ_N
PCIE_WAKE_N S3 D3 M2_PCIE_WAKE_N
FCPO_CTRL_N S4 D4 M2_FCPO_N
U34 TMUX1511PWR V3V7
USB_GATE SEL1
MODEM_USB_ISOLATED_DP S1
USB_DP D1
USB_GATE SEL2
MODEM_USB_ISOLATED_DN S2
USB_DN D2
M2_RESET_N D3
OOB_MODEM_RESET SEL3
OOB_ADC_MODEM_CURRENT D4
MODEM_IMON_TO_ISOLATOR S4
AON_PG SEL4
S3
U35 TLV7011DBVR AON_3V3
USB_VBUS_SENSE IN+
OOB_QUAL_REF IN-
OUT OOB_USB_EXT_VALID
U36 SN74LVC1G08DBVR V3V3
MODEM_START_OK A
B OOB_USB_SESSION_VALID
Y USB_GATE
U7 TPD4E1U06DBVR
SIM1_RST IO1
UIM1_DET IO2
SIM1_CLK IO3
SIM1_IO IO4
NC
U8 ST4SIM-300M SGP.32 experimental population
GND GND
SWIO
ESIM_IO I_O
NC4
VCC UIM2_VDD
RESET ESIM_RST
CLK ESIM_CLK
NC5
U9 USBLC6-2P6
USB_PORT_DP IO1
USB_PORT_DN IO2
USB_PROTECT_DN IO2
VBUS_C VBUS
USB_PROTECT_DP IO1

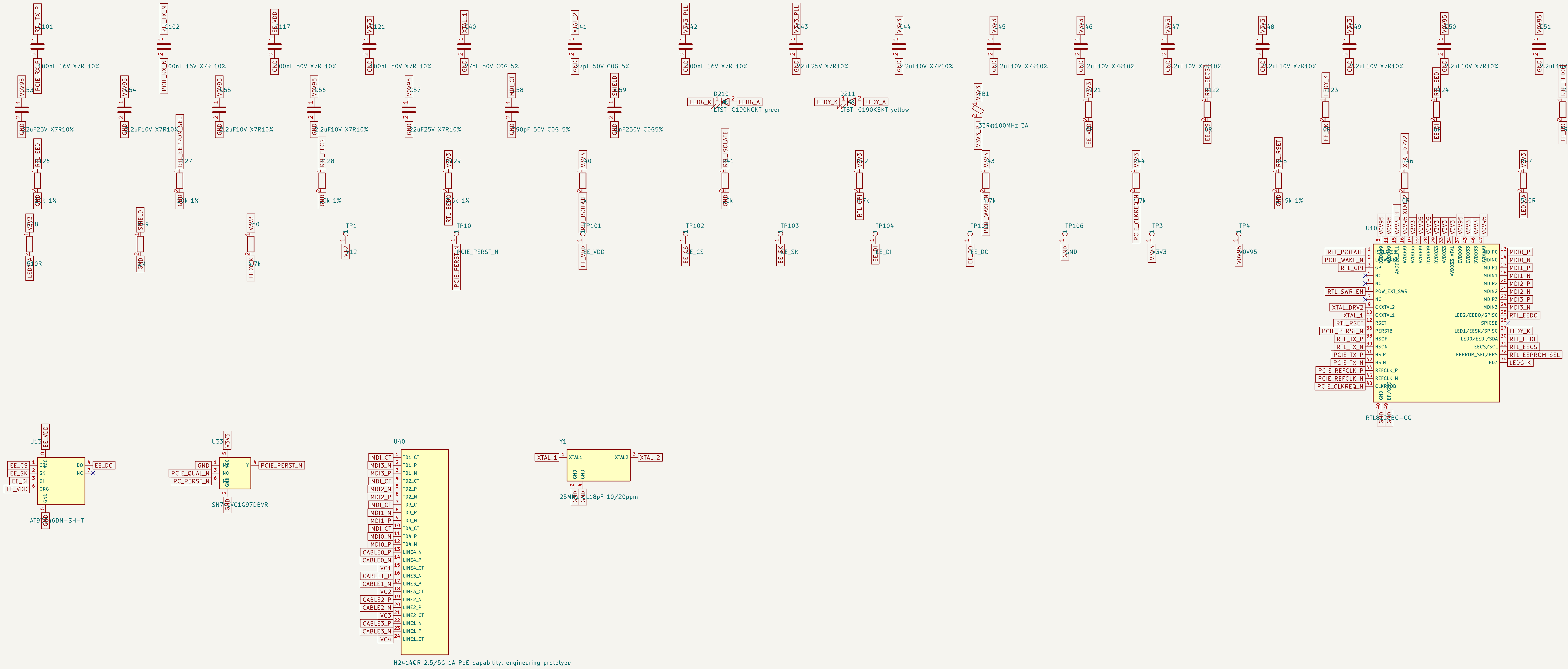
Global labels link sheets; cable return reaches system GND through explicit CMC
Rev A build preparation – current source/native release checks required

Sheet: /RM520N–GL primary / RM551E–GL compatible M.2 socket, SIM 1, eUICC (SIM 2), USB–C/
File: modem–interfaces.kicad_sch

Title: RM520N–GL primary / RM551E–GL compatible M.2 socket, SIM 1, eUIC

Size: A2
Date: 2026–10–03
Rev: A
KiCad E.D.A. 10.0.6
Id: 4/10

RTL8125BG 2.5 GbE



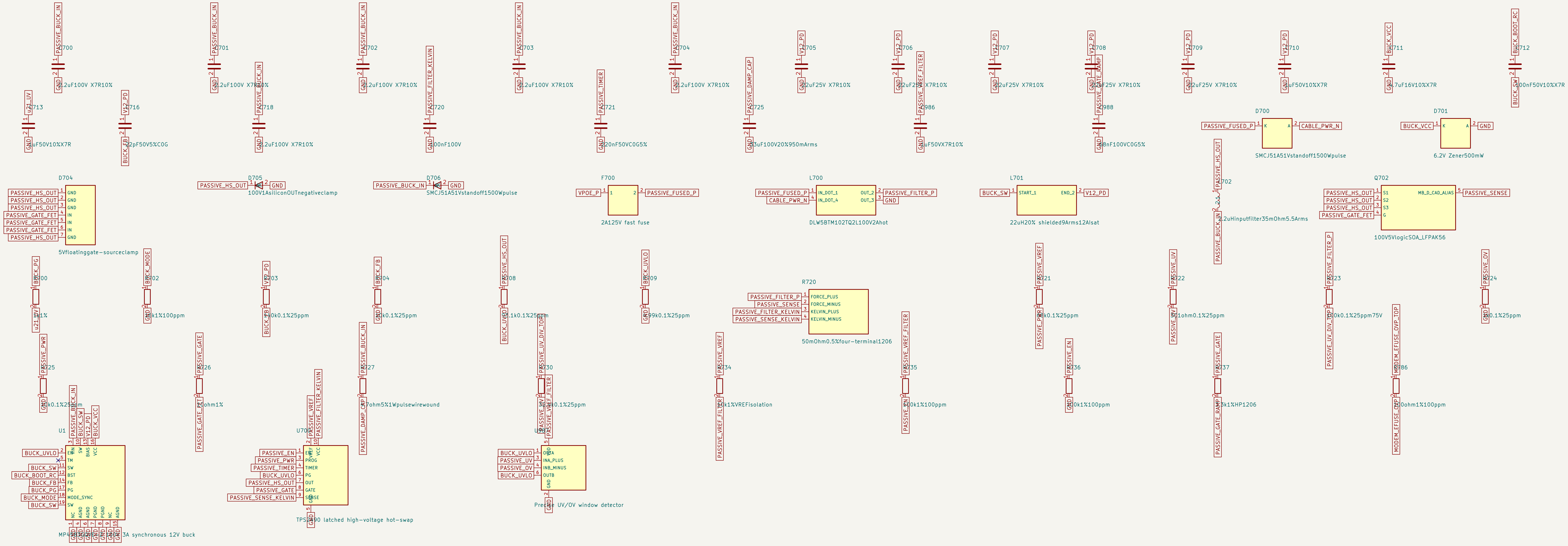
Global labels link sheets; cable return reaches system GND through explicit CMC
Rev A build preparation — current source/native release checks required

Sheet: /RTL8125BG 2.5 GbE/
File: ethernet.kicad_sch

Title: RTL8125BG 2.5 GbE

Size: A2 Date: 2026-10-03 **Rev: A**
KiCad E.D.A. 10.0.6 Id: 5/10

48 V passive input protection and 12 V synchronous buck



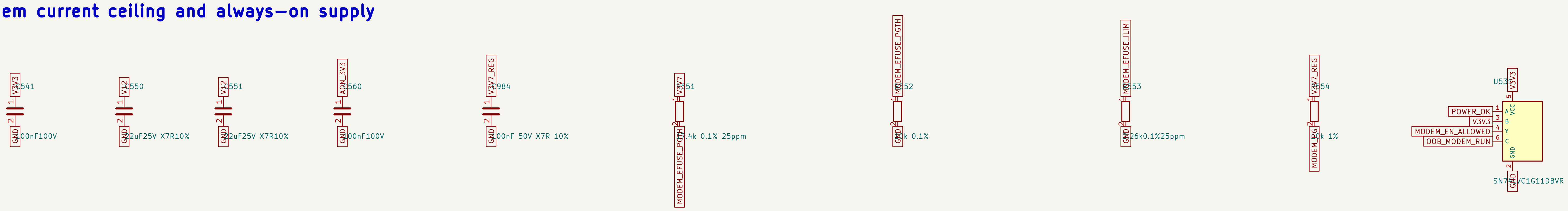
Global labels link sheets; cable return reaches system GND through explicit CMC
Rev A build preparation — current source/native release checks required

Sheet: /48 V passive input protection and 12 V synchronous buck/
File: class6−converter.kicad_sch

Title: 48 V passive input protection and 12 V synchronous buck

Size: A2	Date: 2026−10−03	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 6/10

Modem current ceiling and always-on supply



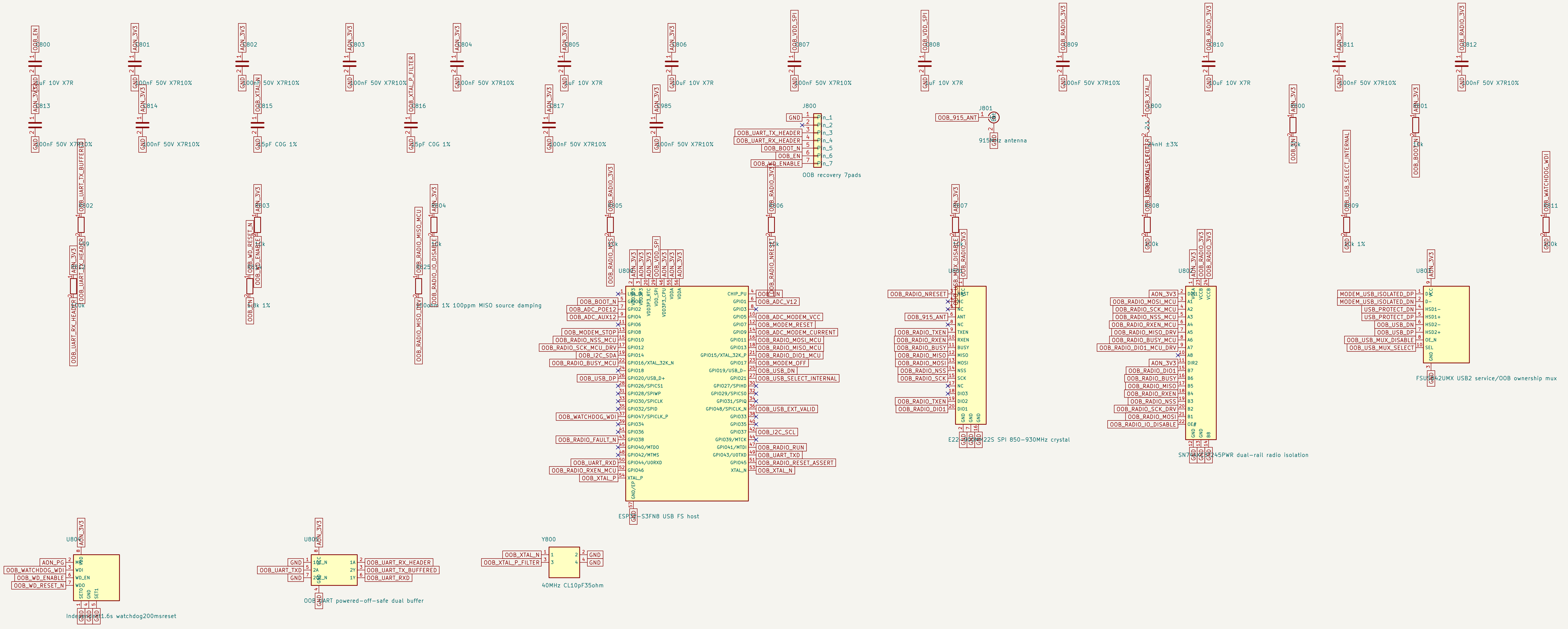
Global labels link sheets; cable return reaches system GND through explicit CMC
Rev A build preparation — current source/native release checks required

Sheet: /Modem current ceiling and always—on supply/
File: class6—grants.kicad_sch

Title: Modem current ceiling and always—on supply

Size: A2	Date: 2026—10—03	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 7/10

OOB controller, US915 radio and recovery interface



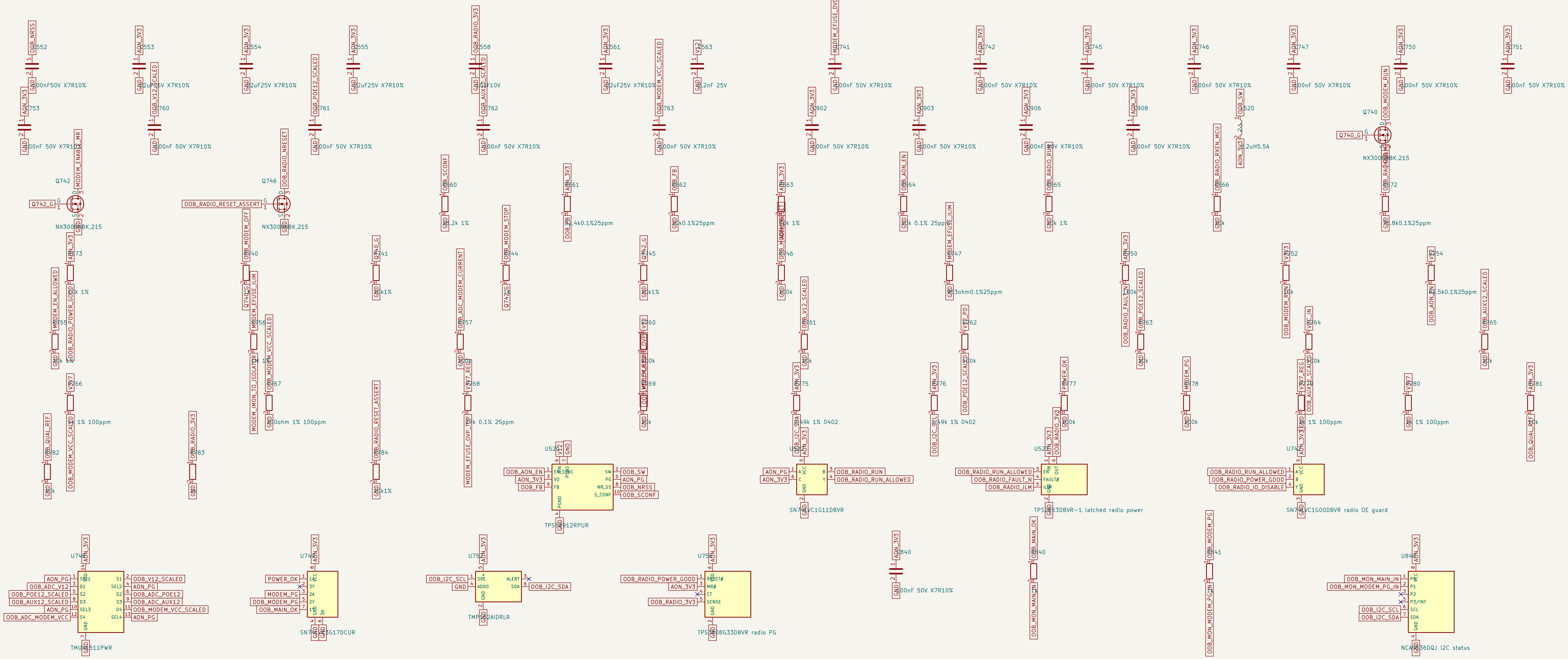
Global labels link sheets; cable return reaches system GND through explicit CMC
Rev A build preparation – current source/native release checks required

Sheet: /OOB controller, US915 radio and recovery interface/
File: oob-controller-radio.kicad_sch

Title: OOB controller, US915 radio and recovery interface

Size: A2	Date: 2026-10-03	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 8/10

OOB monitoring, power control and modem recovery



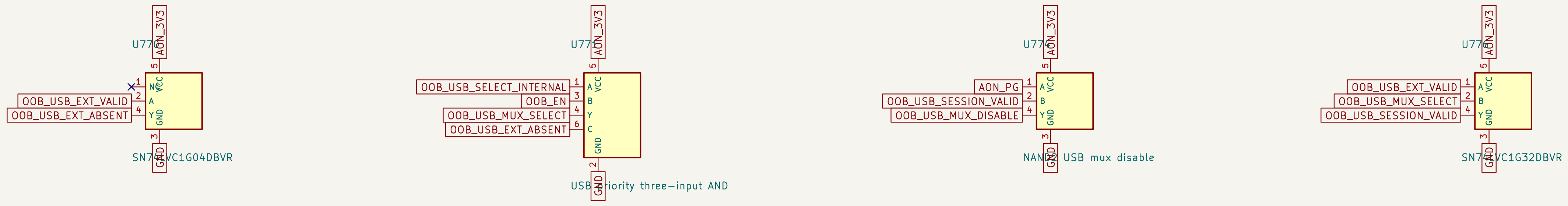
Global labels link sheets; cable return reaches system GND through explicit CMC
Rev A build preparation — current source/native release checks required

Sheet: /OOB monitoring, power control and modem recovery/
File: oob-management.kicad_sch

Title: OOB monitoring, power control and modem recovery

Size: A2 Date: 2026-10-03 **Rev: A**
KiCad E.D.A. 10.0.6 Id: 9/10

OOB USB ownership and reset fallback



Global labels link sheets; cable return reaches system GND through explicit CMC
Rev A build preparation — current source/native release checks required

Sheet: /OOB USB ownership and reset fallback/
File: oob-usb-ownership.kicad_sch

Title: OOB USB ownership and reset fallback

Size: A2
Date: 2026-10-03
Rev: A
KiCad E.D.A. 10.0.6
Id: 10/10